

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AD1-085***

North Shenandoah – Stanley 34.5 kV

August 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Urban Grid Solar Projects, LLC (“Interconnection Customer”) has proposed a new solar generation facility located on Dam Acres Road in Stanley, Page County, Virginia. The requested Maximum Facility Output is 20 MWs with 12.3 MW being recognized by PJM as Capacity Interconnection Rights (CIR). The proposed in-service date for this project is December 31, 2020. **This study does not imply a Potomac Edison (“Transmission Owner”) commitment to this in-service date.**

Point of Interconnection (“POI”)

This project will interconnect with the Potomac Edison transmission system through the Shenandoah Valley Electric Company (or “SVEC”) distribution system at the North Shenandoah substation. This substation is owned by SVEC however, the 138 kV side is owned by Potomac Edison. The Point of Interconnection (POI) will be located at the 138 kV side of the substation.

Interconnection Customer’s facilities will interconnect with the SVEC 34.5 kV distribution system at a point located approximately 3.82 miles from North Shenandoah substation and 3.65 miles from Stanley substation. This interconnection point is the physical “Service Point” or point of common coupling. Please refer to the one-line diagram in Appendix 2 for more details.

Transmission Owner Scope of Work and Costs Summary

The following upgrades are required to support AD1-085 Interconnection:

- (a) Attachment Facilities: None.
- (b) Direct Connection Network Upgrades: None.
- (c) Non-Direct Network Upgrades:
 - (c1) Adjust remote relay and metering settings at North Shenandoah 138 kV Substation. Estimated construction time: six (6) months.
Estimated cost:**\$12,400**
- (d) Direct Local Network Upgrades: None.
- (e) Non-Direct Local Network Upgrades: None.
- (f) Option to Build Upgrades: None.

Estimated Total Costs (a) to (f):\$12,400

NOTE: The above shown estimated costs do not include Contribution in Aid of Construction (CIAC) Federal Income Tax Gross Up charge. The tax Dollars may or may not be charged to this project depending upon whether this project meets the eligibility requirements of the latest IRS Safe Harbor provisions for non-taxable status.

Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FirstEnergy's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the Interconnection Customer is responsible for the following:

1. The purchase and installation of fully rated 34.5 kV circuit breaker to protect the AD1-085 generator lead line; A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FirstEnergy generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (SCADA) equipment to provide information in a compatible format to the FirstEnergy Transmission System Control Center.
4. A compliance with the FirstEnergy and PJM generator power factor and voltage control requirements.

5. The execution of a back-up service agreement to serve the customer load supplied from the AD1-085 generation project metering point when the units are out-of-service. This assumes the intent of Interconnection Customer is to net the generation with the load.
6. Power Factor Requirements: Interconnection Customer shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FirstEnergy transmission system.
7. System Protection Requirements: Interconnection Customer must design its Customer Facilities in accordance with all applicable standards, including the standards in FirstEnergy's "Requirements for Transmission Connected Facilities" document located at above shown link. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.
8. Revenue Metering and SCADA Requirements: PJM requires that Interconnection Customer must install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for Interconnection Customer's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2. FirstEnergy requires that Interconnection Customer and SVEC must comply with all FirstEnergy Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "FirstEnergy Requirements for Transmission Connected Facilities" document located at the following links:
<http://www.firstenergycorp.com/feconnect> and <http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>
9. Interconnection Customer must meet all PJM, ReliabilityFirst and NERC reliability criteria and operating procedures required for standards compliance. For example, Interconnection Customer will need to properly locate and report the over and under-voltage and over and under-frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FirstEnergy system.
10. Interconnection Customer will be responsible for constructing all of the facilities on its side of the POI including all facilities associated with the attachment line and the AD1-085 point of common coupling to SVEC system. Interconnection Customer will be responsible for acquiring all easements, properties and permits that may be required to construct their line and the associated attachment facilities. Interconnection Customer may not install above ground equipment within any Transmission Owner's right-of-way unless permission to do so is expressly granted by the Transmission Owner.
11. If Interconnection Customer's intention is to participate at the PJM wholesale electricity market, then they must enter into a separate two-party interconnection agreement with the Shenandoah Valley Electric Cooperative prior to the execution of a Wholesale Market Participation Agreement with PJM and FirstEnergy.

The above requirements are in addition to any metering or other requirements imposed by PJM and/or SVEC.

Network Impacts

The Queue Project AD1-085 was evaluated as a 20.0 MW (Capacity 12.3 MW) injection into the North Shenandoah 138 kV substation in the APS area. Project AD1-085 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD1-085 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Short Circuit

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis - 2021

Not required.

System Reinforcements

Short Circuit

None

Stability and Reactive Power Requirement

None

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Winter Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Winter Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

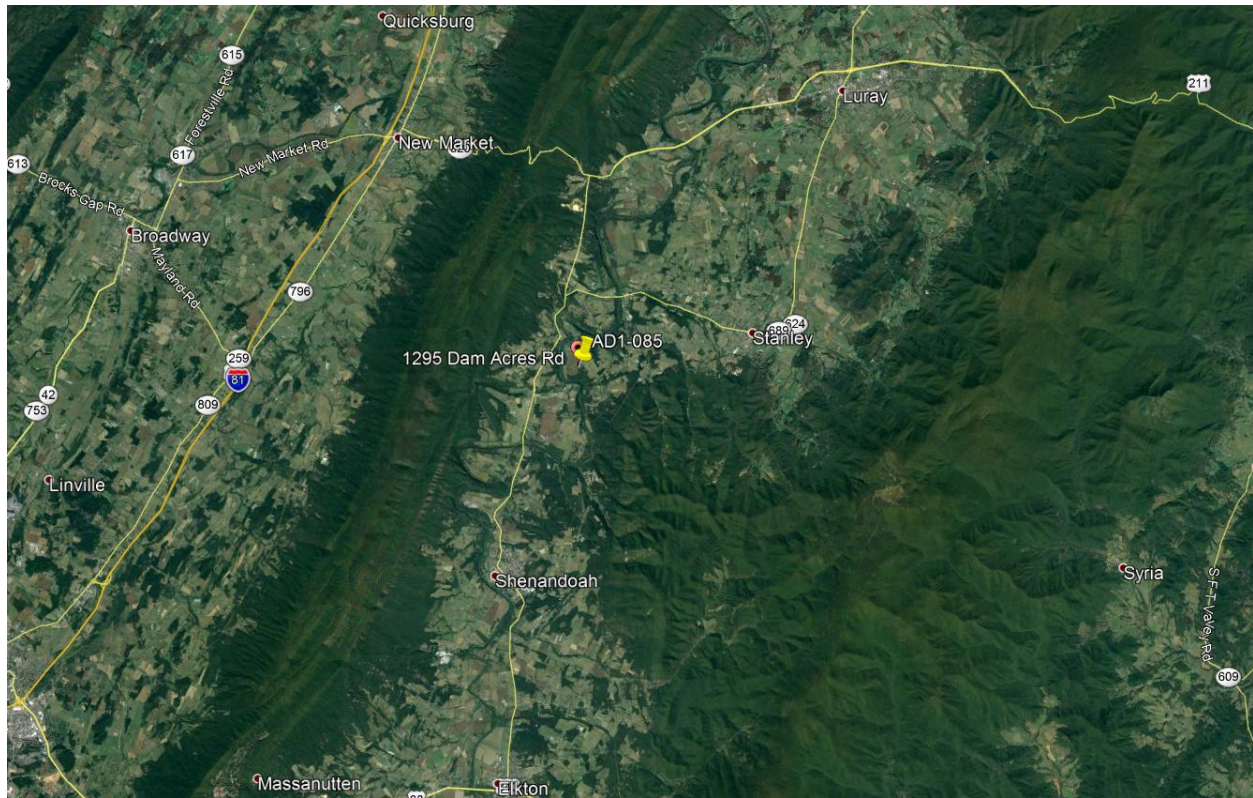
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Appendix 1

Facility Location

PJM Queue Position: AD1-085



Appendix 2

Interconnection One-Line Diagram

PJM Queue Position: AD1-085

North Shenandoah 138 kV Substation via SVEC North Shenandoah – Stanley 34.5 kV Line

